

Cost-Related Barriers to Care and Medications Among US Stroke Survivors, 2019–2025

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Abstract

Background: Secondary prevention after stroke requires continued access to follow-up care and medications. Recent national estimates for stroke survivors have addressed medication costs only. We estimated the prevalence of cost-related barriers to both medical care and medications among US stroke survivors from 2019 through 2025 and examined which groups were most affected.

Methods: We conducted a repeated cross-sectional analysis of the 2019–2025 National Health Interview Survey. The primary outcome was any past-year cost-related delay in or nonreceipt of medical care, nonreceipt of needed prescription medication, or medication underuse to save money. Survey-weighted modified Poisson models with sequential adjustment estimated prevalence ratios (PRs), incorporating all ten National Center for Health Statistics income imputations; logistic models estimated covariate-standardized annual prevalence.

Results: Among 7,104 stroke survivors, 1,184 reported a cost-related barrier, a weighted prevalence of 17.9% (95% CI, 16.8%–19.1%). Prevalence was 28.0% (95% CI, 25.9%–30.2%) at ages 18–64 years versus 10.9% (95% CI, 9.8%–12.0%) at 65 years and older, and 58.7% (95% CI, 50.7%–66.3%) among uninsured survivors. After full adjustment, working-age survivors remained more than twice as likely to report a barrier (PR, 2.18; 95% CI, 1.90–2.50). Income of 100%–199% of the federal poverty level (PR, 2.50; 95% CI, 1.94–3.21), uninsured status (PR, 1.79; 95% CI, 1.49–2.16), and disability (PR, 1.19; 95% CI, 1.05–1.35) were independently associated with barriers. Standardized prevalence was 19.2% (95% CI, 16.6%–22.0%) in 2019 and 17.0% (95% CI, 14.3%–20.1%) in 2025, with no evidence of change across years ($P=0.258$).

Conclusions: About 1 in 6 US stroke survivors, and more than 1 in 4 of those aged 18–64 years, reported delaying or forgoing care or medications because of cost, and prevalence did not decline

between 2019 and 2025. Affordability should be assessed routinely during post-stroke follow-up, particularly among working-age, lower-income, and uninsured survivors.

Non-standard Abbreviations and Acronyms

FPL, federal poverty level; NCHS, National Center for Health Statistics; NHIS, National Health Interview Survey; PR, prevalence ratio; STROBE, Strengthening the Reporting of Observational Studies in Epidemiology

Introduction

Millions of US adults have survived a stroke and remain at elevated risk of recurrent vascular events.[1] Guidelines for secondary prevention recommend long-term control of blood pressure, lipids, and glucose and continued antithrombotic therapy,[2] which require regular outpatient follow-up and uninterrupted access to prescription medications. Out-of-pocket costs can limit both.

Cost-related barriers after stroke have been described for more than 2 decades. In the 1998–2002 National Health Interview Survey (NHIS), younger stroke survivors were markedly less likely than older survivors to obtain physician care and medications,[3] and NHIS data from 1999–2010 documented substantial cost-related medication nonadherence that varied with age and insurance.[4] Financial stress and food insecurity were linked to medication nonadherence among stroke survivors in 2014–2018,[5] and a recent analysis of the 2020–2023 NHIS reported that 13% of stroke survivors had cost-related medication nonadherence, most often those who were younger, uninsured, or of lower income.[6] Among working-age US adults with multiple chronic conditions, cost-related nonadherence declined between 2019 and 2023.[7]

Recent stroke-specific studies have examined medication nonadherence but not delayed or forgone medical care, which affects the visits at which preventive therapy is prescribed and adjusted. No study has covered the full period from 2019, the first year of the redesigned NHIS, through 2025, which included the COVID-19 pandemic and temporary coverage protections. The contribution of income, insurance, and health status to the higher prevalence among working-age survivors has not been quantified.

We used the 2019–2025 NHIS to estimate the prevalence of cost-related barriers to medical care and medications among US stroke survivors, describe their variation across years, and quantify

their distribution by age, income, insurance, and disability. We also examined how the association with working age changed under sequential adjustment for socioeconomic and clinical characteristics.

Methods

Data availability

NHIS public-use files and documentation are available from NCHS. All acquisition, analysis, verification, and manuscript-generation code, source file hashes, aggregate results, tables, and figures are publicly available at <https://github.com/blakepi/nhis-stroke-affordability> (release v1.2.0). One author (G.B.P.) had full access to all the data in the study and takes responsibility for its integrity and the data analysis.

Data source and study population

NHIS is a continuous, cross-sectional household survey of the civilian noninstitutionalized US population conducted by the National Center for Health Statistics (NCHS), with a stratified, multistage probability design.[8] We conducted a repeated cross-sectional analysis of the public-use Sample Adult files for 2019 through 2025. Stroke survivors were adults who reported ever having been told by a doctor or other health professional that they had a stroke. We required a positive survey weight and a valid age of 18–85 years (85 representing 85 years and older); 10 records with unknown age were excluded.

Annual estimates used each year's full Sample Adult file and its annual weight. Because the 2020 survey re-interviewed a subset of 2019 respondents, pooled analyses followed NCHS guidance for combining 2020 with other years: the follow-back respondents were omitted, and the remaining 2020 records were assigned the partial-sample weight.[9] Weights were divided by

seven to represent an average annual population. The public-use stratum and primary sampling unit identifiers were used without modification, consistent with NCHS instructions for pooling survey years from 2019 onward.[8] Survey designs were specified on the full adult files before restricting analyses to the stroke-survivor domain, preserving the design structure for variance estimation. Because NHIS data are publicly available and de-identified, this study did not require institutional review board approval or informed consent.

Outcomes

The primary outcome was any cost-related barrier in the preceding 12 months, defined from six questions. Three questions asked of all adults captured delaying medical care because of cost, needing but not getting medical care because of cost, and needing but not getting prescription medication because of cost. Three questions asked of adults who had been prescribed medication in the past 12 months captured skipping doses, taking less medication, and delaying a prescription fill to save money.

A survivor with a "yes" response to any applicable question was classified as having a barrier. Classification as having no barrier required "no" responses to all three universal questions and either no prescription use or "no" responses to all three medication-use questions. Other response patterns were classified as unknown. Secondary outcomes were each component separately, medication underuse (any of the three medication-use questions, among adults prescribed medication), and a three-question composite restricted to the universal items. Question wording and universes were verified against each year's codebook (Supplemental Methods; Tables S1 and S2).

Covariates

Demographic covariates were survey year, age (natural cubic spline with three degrees of freedom), sex, race and Hispanic origin (seven published categories), and Census region.

Socioeconomic covariates were education (less than high school; high school or equivalent; some college or associate degree; bachelor's degree or higher), family income as a percentage of the federal poverty level (FPL; below 100%, 100%–199%, 200%–399%, and 400% or higher), and health insurance (private coverage with or without public coverage; public coverage without private coverage; military coverage only; uninsured). Clinical and health-status covariates were diagnosed hypertension, diabetes, and coronary heart disease; cigarette smoking (never, former, current); fair or poor self-rated health; and disability, defined by the NCHS recode of the Washington Group questions as substantial difficulty in at least one functional domain. The education variable changed name between 2020 and 2021 and was harmonized across years.

Statistical analysis

Prevalence estimates were survey-weighted with Taylor-series linearization for variance.[10]

Annual confidence intervals used a logit transformation; pooled component intervals used a design-adjusted beta method, and subgroup estimates were pooled on the logit scale. All reported prevalence estimates met the NCHS presentation standards for proportions.[11]

Family income was imputed by NCHS for 26.8% of stroke survivors, and NCHS releases ten multiply imputed income files per year.[12] All analyses involving income were repeated in each of the ten completed data sets and combined with Rubin's rules using finite design degrees of freedom. Other covariates had little missingness (largest, 3.5% for smoking) and were handled by complete-case analysis.

Survey-weighted modified Poisson regression with robust variance estimated prevalence ratios for the primary outcome.[10,13] Three nested models were fit in the same analytic sample: Model 1 included year and demographic covariates; Model 2 added education, income, and insurance; and Model 3 added the clinical and health-status covariates. A parallel set of models replaced the age spline with a working-age indicator (18–64 versus 65 years and older) to quantify attenuation of the age-group association across adjustment sets. Adjusted annual prevalence was estimated by fitting a logistic model with the Model 3 covariates and standardizing each year's prediction to the pooled covariate distribution,[14] with variance accounting for both model and reference-distribution uncertainty. We tested categorical year and a year-by-age-group interaction using pooled Wald F tests.

Sensitivity analyses excluded 2020, restricted the sample to working-age adults or to adults without disability, modeled each component outcome, and added explicit "unknown" categories for missing covariates. Because modified Poisson models can yield fitted values above one, key contrasts were re-estimated as standardized prevalence ratios from bounded logistic models. All primary prevalence estimates, standard errors, and the full model were independently reproduced in a second software environment. Analyses used R version 4.6.0 with the survey and mitools packages. P values are two-sided without adjustment for multiplicity. Reporting follows the STROBE guideline;[15] the completed checklist is provided in the Supplemental Material.

Results

Study population

The seven annual files contained 207,064 sample adults, of whom 7,565 reported a prior stroke (Figure S1; Table S3). After exclusion of 10 records with unknown age and application of the pooled-sample rules for 2020, the pooled sample comprised 7,181 stroke survivors, of whom

7,104 (98.9%) had a classifiable primary outcome; 6,777 had complete covariates for adjusted models (Table S4). In the weighted pooled sample, 41.0% were aged 18–64 years, 51.5% were female, 38.8% had a disability, and 51.9% rated their health as fair or poor (Table 1 and Table S5). More than half (53.6%) had public coverage without private insurance, 40.3% had private coverage, and 4.0% were uninsured.

Prevalence of cost-related barriers

Overall, 1,184 survivors reported at least one cost-related barrier, a weighted prevalence of 17.9% (95% CI, 16.8%–19.1%), or approximately 1 in 6 stroke survivors (Table 2). The most common barrier was needing but not getting a prescription medication because of cost (10.0%; 95% CI, 9.1%–11.0%), followed by delaying medical care (8.5%; 95% CI, 7.7%–9.4%) and forgoing medical care (8.4%; 95% CI, 7.6%–9.3%) (Figure 1B). Among survivors prescribed medication, 10.1% (95% CI, 9.2%–11.1%) reported skipping doses, taking less, or delaying fills to save money. Restricting the composite to the three universal questions gave a prevalence of 15.5% (95% CI, 14.3%–16.6%) (Table S6).

Observed annual prevalence ranged from 15.3% to 20.6% without a monotonic trend (Figure 1A; Table 2). After standardization for demographic, socioeconomic, and clinical characteristics, prevalence was 19.2% (95% CI, 16.6%–22.0%) in 2019 and 17.0% (95% CI, 14.3%–20.1%) in 2025, an absolute difference of –2.1 percentage points (95% CI, –6.0 to 1.8). Prevalence did not differ significantly by year ($P=0.258$), and the age-group difference was stable over time ($P=0.830$ for interaction).

Who reports cost-related barriers

Prevalence differed by age, income, and insurance (Figure 2; Table S7). Working-age survivors reported barriers at 28.0% (95% CI, 25.9%–30.2%), compared with 10.9% (95% CI, 9.8%–

12.0%) among survivors aged 65 years and older. Prevalence was higher at lower income levels: 8.2% at 400% FPL or higher, 18.0% at 200%–399% FPL, 23.5% at 100%–199% FPL, and 22.5% below 100% FPL. A barrier was reported by 58.7% (95% CI, 50.7%–66.3%) of uninsured survivors, compared with 16.6% (95% CI, 14.9%–18.4%) of privately insured survivors and 15.8% (95% CI, 14.5%–17.3%) of those with public coverage only. Survivors with a disability reported barriers more often than those without: 20.2% (95% CI, 18.4%–22.1%) versus 16.5% (95% CI, 15.1%–17.9%).

In adjusted models, the association with working age was attenuated but persisted (Table 3). The prevalence ratio for ages 18–64 versus 65 years and older was 2.51 (95% CI, 2.20–2.87) with demographic adjustment, 2.27 (95% CI, 1.98–2.61) after adding education, income, and insurance, and 2.18 (95% CI, 1.90–2.50) after further adjustment for clinical and health-status characteristics. Relative to income of 400% FPL or higher, fully adjusted prevalence ratios were 2.07 (95% CI, 1.65–2.61) for 200%–399% FPL, 2.50 (95% CI, 1.94–3.21) for 100%–199% FPL, and 2.00 (95% CI, 1.52–2.63) for below 100% FPL. Uninsured survivors had a prevalence ratio of 1.79 (95% CI, 1.49–2.16) relative to privately insured survivors, whereas public coverage without private insurance was associated with a lower adjusted prevalence (PR, 0.72; 95% CI, 0.63–0.83). Disability was associated with a prevalence ratio of 1.19 (95% CI, 1.05–1.35). Female sex (PR, 1.27; 95% CI, 1.12–1.43), coronary heart disease (PR, 1.37; 95% CI, 1.18–1.58), current smoking (PR, 1.21; 95% CI, 1.04–1.42), and fair or poor self-rated health (PR, 1.58; 95% CI, 1.37–1.82) were also independently associated with barriers (Table S8).

Sensitivity analyses

Results were similar when 2020 was excluded (income 100%–199% FPL: PR, 2.59; 95% CI, 2.00–3.36; uninsured: PR, 1.82; 95% CI, 1.49–2.21; disability: PR, 1.18; 95% CI, 1.04–1.34) and

were consistent in direction across component outcomes and the missing-category analysis (Table S9). Within working-age survivors, the disability association was smaller (PR, 1.08; 95% CI, 0.92–1.27). The fully adjusted modified Poisson model produced fitted values above one for approximately 0.2% of records; standardized prevalence ratios from bounded logistic models were similar: 2.13 (95% CI, 1.86–2.44) for working age, 2.41 (95% CI, 1.88–3.09) for income 100%–199% FPL, 1.99 (95% CI, 1.63–2.43) for uninsured status, and 1.18 (95% CI, 1.04–1.33) for disability (Table S10). Model convergence and fitted-value diagnostics are given in Table S11. Barrier prevalence among the 327 survivors excluded from adjusted models for missing covariates (19.2%; 95% CI, 14.0%–25.5%) was similar to that in the analytic sample (17.8%).

Discussion

In this nationally representative analysis spanning 2019 through 2025, approximately 1 in 6 US stroke survivors reported delaying or forgoing medical care, going without a needed prescription, or rationing medication because of cost in the preceding year. More than 1 in 4 working-age survivors and nearly 3 in 5 uninsured survivors reported a barrier, prevalence was higher at lower income levels, and survivors with disability were affected more often than those without.

Working-age survivors remained more than twice as likely as older survivors to report a barrier after adjustment for income, insurance, education, and health status. Prevalence did not decline between 2019 and 2025.

These findings are consistent with earlier NHIS studies, in which younger stroke survivors were less able to obtain physician care and medications in 1998–2002[3] and cost-related medication nonadherence was common through 2010.[4] The most recent stroke-specific estimate, 13% cost-related medication nonadherence in 2020–2023,[6] is consistent with the 10.1% (95% CI, 9.2%–11.1%) medication underuse and 10.0% (95% CI, 9.1%–11.0%) forgone prescriptions

observed here. Our estimates add delayed and forgone medical care, each reported by approximately 8% of survivors, showing that cost limits outpatient visits as well as medication use. The absence of a decline across 2019–2025 differs from the decrease in cost-related nonadherence reported among working-age adults with multiple chronic conditions over 2019–2023.[7]

Medicare eligibility at 65 years is a frequently cited explanation for age differences in affordability. The prevalence ratio for working age decreased from 2.51 to 2.18 (95% CI, 1.90–2.50) across models, so income and insurance explained part of the difference, but most of the excess remained after adjustment. Unlike Medicare beneficiaries, most working-age survivors rely on employer-sponsored or individually purchased coverage with deductibles and cost sharing, and a stroke can end both employment and the coverage attached to it. Underinsurance, which NHIS does not measure, is a likely contributor among privately insured survivors, 16.6% of whom reported a barrier. Affordability screening should therefore extend beyond uninsured patients.

Survivors with public coverage had lower adjusted prevalence than those with private coverage. Public coverage in this population is mostly Medicare, often combined with Medicaid or supplemental coverage that limits out-of-pocket costs. The association with disability persisted after adjustment, consistent with the greater care needs and reduced earning capacity that follow disabling stroke.

Cost-related nonadherence after stroke is common,[4,6] and identifying it requires asking patients directly. Brief affordability screening at follow-up, focused on survivors younger than 65 years, those with lower income, and those with disability, identifies candidates for generic substitution, patient assistance programs, insurance navigation, and social work referral. Cost-

related barriers in this population did not decline under the coverage policies in effect during 2019–2025.

Strengths and limitations

Strengths include seven consecutive years of nationally representative data through 2025, a multidomain outcome that captures barriers to visits as well as medications, verification of question wording and universes across years, use of the NCHS-recommended weights for pooling the 2020 survey, incorporation of all ten income imputations, sequential adjustment within a single analytic sample, and independent reproduction of the primary results.

This study has limitations. Stroke history and cost-related barriers were self-reported. The survey does not record stroke subtype, timing, or severity, so some barriers may predate the stroke.

NHIS excludes institutionalized adults, who include the most severely disabled survivors. The composite outcome does not capture the frequency or duration of barriers, whether forgone care was stroke related, or which medications were rationed. Insurance and disability were measured at interview, whereas barriers refer to the preceding year. Wealth, out-of-pocket spending, benefit design, and caregiving support were not measured, and the associations are cross-sectional. Nonincome covariates were analyzed as complete cases. The analysis was not preregistered.

Conclusions

Cost-related barriers to medical care and medications were reported by approximately 1 in 6 US stroke survivors and more than 1 in 4 of those aged 18–64 years, with no decline between 2019 and 2025. Affordability should be assessed routinely during post-stroke follow-up, particularly for younger, lower-income, uninsured, and disabled patients.

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Supplemental Material

Supplemental Methods

Figure S1

Tables S1–S12

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Tables

Table 1. Characteristics of US stroke survivors in the pooled 2019–2025 NHIS sample, overall and by cost-related barrier status

Characteristic	Overall (N=7,104)	No barrier (n=5,920)	Any barrier (n=1,184)
Age 65 years or older	4,763 (59.0)	4,237 (64.0)	526 (35.9)
Age 18–64 years	2,341 (41.0)	1,683 (36.0)	658 (64.1)
Male	3,269 (48.5)	2,777 (49.7)	492 (43.3)
Female	3,835 (51.5)	3,143 (50.3)	692 (56.7)
Income $\geq$ 400% FPL	1,681.2 (24.2)	1,559.9 (27.0)	121.3 (11.0)
Income 200–399% FPL	2,133.3 (31.0)	1,786.9 (31.0)	346.4 (31.2)
Income 100–199% FPL	1,986.9 (27.5)	1,549.6 (25.6)	437.3 (36.0)
Income <100% FPL	1,302.6 (17.4)	1,023.6 (16.4)	279.0 (21.8)
Private, with or without public coverage	2,888 (40.3)	2,465 (40.9)	423 (37.2)
Public without private	3,871 (53.6)	3,263 (54.9)	608 (47.4)
Military only	109 (1.6)	92 (1.6)	17 (1.7)
Uninsured	207 (4.0)	80 (2.0)	127 (13.2)
Insurance unknown	29 (0.5)	20 (0.5)	9 (0.6)
Without disability	4,384 (61.2)	3,742 (62.3)	642 (56.3)
With disability	2,720 (38.8)	2,178 (37.7)	542 (43.7)
Good or better self-rated health	3,607 (48.0)	3,201 (51.6)	406 (31.7)
Fair or poor self-rated health	3,486 (51.9)	2,712 (48.3)	774 (68.1)
Self-rated health unknown	11 (0.1)	7 (0.1)	4 (0.2)
Never smoked	3,196 (46.0)	2,710 (46.6)	486 (43.5)
Former smoker	2,592 (34.0)	2,240 (35.5)	352 (27.2)
Current smoker	1,132 (16.8)	821 (14.7)	311 (26.3)
Smoking unknown	184 (3.2)	149 (3.2)	35 (3.0)

Values are unweighted number (survey-weighted column percentage). Income counts are averaged across the ten NCHS imputations and may be fractional. Unknown values are retained in denominators; percentages within a variable may not sum to 100 because of rounding. FPL indicates federal poverty level; NHIS, National Health Interview Survey. Additional characteristics are given in Table S5.

Table 2. Observed and standardized annual prevalence of any cost-related barrier among US stroke survivors, 2019–2025

Year	No. with observed outcome	No. with barrier	Observed prevalence, % (95% CI)	Standardized prevalence, % (95% CI)
2019	1,191	231	20.6 (17.9–23.6)	19.2 (16.6–22.0)
2020	1,051	156	17.3 (14.4–20.6)	17.1 (13.8–21.0)
2021	1,001	179	19.4 (16.5–22.8)	18.5 (15.6–21.8)
2022	999	147	16.0 (13.4–19.0)	16.0 (13.6–18.8)
2023	1,116	169	15.3 (12.8–18.2)	16.4 (13.9–19.2)
2024	1,229	215	20.0 (17.3–23.1)	20.4 (17.8–23.4)
2025	889	138	16.8 (14.0–20.0)	17.0 (14.3–20.1)

Observed estimates use each year's full sample and annual weights. Standardized estimates come from the fully adjusted logistic model fitted in the pooled complete-case sample with 2020 partial-sample weights and ten income imputations, standardized to the pooled covariate distribution. Confidence intervals account for the complex survey design.

Table 3. Prevalence ratios for any cost-related barrier among US stroke survivors under sequential adjustment

Characteristic	Model 1, PR (95% CI)	Model 2, PR (95% CI)	Model 3, PR (95% CI)
Age 18–64 vs ≥65 years	2.51 (2.20–2.87)	2.27 (1.98–2.61)	2.18 (1.90–2.50)
Female vs male	1.30 (1.14–1.47)	1.25 (1.10–1.42)	1.27 (1.12–1.43)
Some college/associate vs bachelor's or higher	—	0.94 (0.79–1.12)	0.89 (0.75–1.06)
High school/GED vs bachelor's or higher	—	0.81 (0.67–0.97)	0.75 (0.62–0.89)
Less than high school vs bachelor's or higher	—	0.91 (0.73–1.13)	0.80 (0.64–0.99)
Income 200–399% vs ≥400% FPL	—	2.29 (1.81–2.89)	2.07 (1.65–2.61)
Income 100–199% vs ≥400% FPL	—	2.95 (2.30–3.78)	2.50 (1.94–3.21)
Income <100% vs ≥400% FPL	—	2.43 (1.86–3.19)	2.00 (1.52–2.63)
Public without private vs private	—	0.79 (0.68–0.91)	0.72 (0.63–0.83)
Military only vs private	—	0.83 (0.53–1.30)	0.81 (0.52–1.28)
Uninsured vs private	—	1.76 (1.46–2.12)	1.79 (1.49–2.16)
Hypertension vs none	—	—	0.99 (0.86–1.14)
Diabetes vs none	—	—	1.12 (0.97–1.30)
Coronary heart disease vs none	—	—	1.37 (1.18–1.58)
Former vs never smoking	—	—	0.98 (0.85–1.14)
Current vs never smoking	—	—	1.21 (1.04–1.42)
Fair/poor vs good or better health	—	—	1.58 (1.37–1.82)
Disability vs none	—	—	1.19 (1.05–1.35)

Survey-weighted modified Poisson regression with ten income imputations; all models include the same 6,777 survivors (1,117 with a barrier). Model 1 adjusts for survey year, age (spline), sex, race and Hispanic origin, and region. Model 2 adds education, family income, and insurance. Model 3 adds hypertension, diabetes, coronary heart disease, smoking, self-rated health, and disability. The age-group estimate is from a parallel model that replaces the age spline with an age-group indicator. Private coverage (with or without public coverage) is the reference for insurance. A dash indicates that the variable was not in the model. FPL indicates federal poverty level; GED, General Educational Development; PR, prevalence ratio. Complete Model 3 estimates are given in Table S8.

Figures

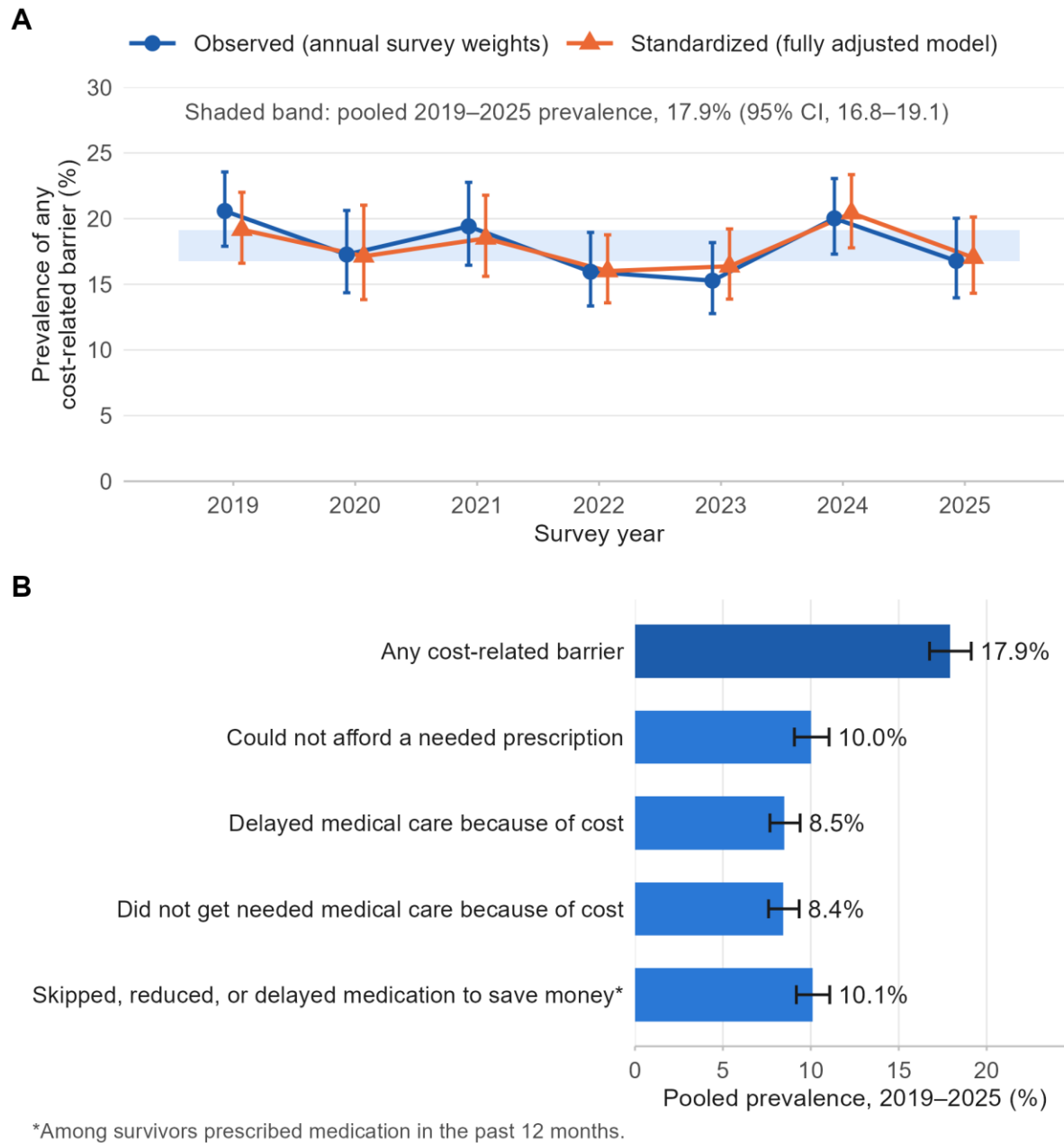


Figure 1. Cost-related barriers among US stroke survivors, 2019–2025. (A) Annual prevalence of any cost-related barrier, observed with annual survey weights (blue circles) and standardized to the pooled covariate distribution from the fully adjusted logistic model (orange triangles). (B) Pooled prevalence of the composite outcome and each component barrier. Medication underuse is estimated among survivors prescribed medication in the past 12 months. Error bars are 95% CIs.

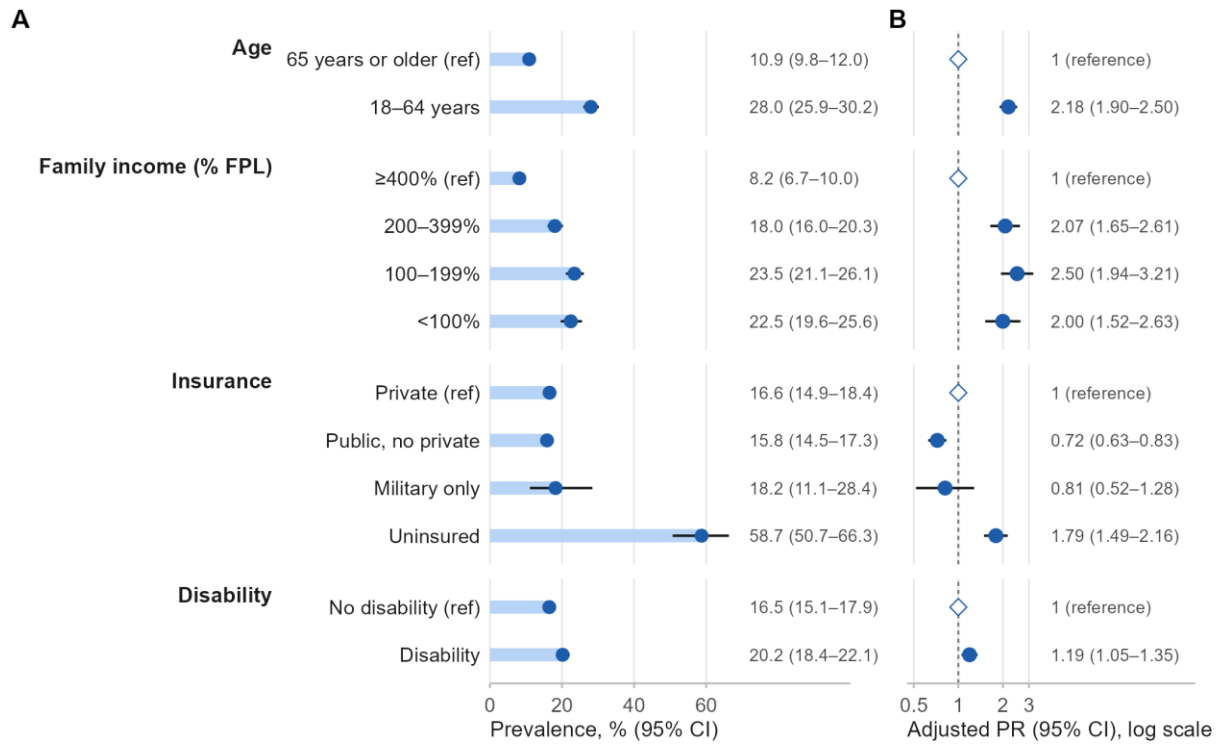


Figure 2. Distribution of cost-related barriers by survivor characteristics. (A) Unadjusted pooled prevalence of any cost-related barrier within groups defined by age, family income, insurance coverage, and disability. (B) Fully adjusted prevalence ratios (Model 3) from survey-weighted modified Poisson regression incorporating ten income imputations; the age contrast is from the parallel model that replaces the age spline with an age-group indicator. Reference groups are age 65 years and older, income 400% or more of the federal poverty level (FPL), private insurance, and no disability. Error bars are 95% CIs; the dashed line marks a prevalence ratio of 1. PR indicates prevalence ratio.